

Comparative Assessment: MRBENCH vs. MATHDIAL

Deployment context: Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)

Deployment Context

Use case: Deployment scenario: A GPT-4 model is implemented in a one-on-one tutoring session, where its responses are augmented with a human tutor’s input. The tutor evaluates the pedagogical quality of the combined response, particularly in terms of how well it provides guidance. The goal is to assess whether the augmented GPT-4 responses are acceptable to the tutor.

Domain: Educational tutoring

Setting: Mobile application / enterprise software

Target population: High-end professional teacher teaching grade 1-8 in major metropolitan cities in India, such as Delhi and Mumbai, who is fluent in English.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
MRBENCH	HIGH	3.2 / 5
MATHDIAL	HIGH	3.0 / 5

Dimension Scores Comparison

Dimension	MRBENCH	Rating	MATHDIAL	Rating	Δ
Input Ontology (IO)	4/5	Minor gaps	4/5	Minor gaps	0
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	4/5	Minor gaps	5/5	Strong alignment	-1
Output Ontology (OO)	3/5	Moderate gaps	2/5	Significant gaps	+1
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	4/5	Minor gaps	5/5	Strong alignment	-1
Average	3.2		3.0		+0.2

Δ = MRBENCH score – MATHDIAL score. Positive = regional benchmark scores higher.

Overall Summaries

MRBENCH (risk: HIGH)

MRBench is a well-constructed, narrowly-scoped benchmark for evaluating AI tutor responses to mathematics mistakes, with strengths in taxonomy validation, annotator training, and metric design. For the Delhi/Mumbai Grade 1–8 mathematics teacher deployment, validity is strongly preserved on dimensions involving curricular ontology (IO), input form (IF), and output form (OF), all of which align with the user’s accepted text-only English, curriculum-agnostic deployment. However, two HIGH-priority dimensions show significant validity concerns: (a) Input Content — Bridge and MathDial were sourced from Western academic settings and depict student communication dynamics (extended verbalization, structured Socratic exchanges) that the user explicitly identified as mismatched with Indian classroom realities; (b) Output Content — gold-standard labels were produced by four MBZUAI CS post-graduates without required teaching experience and without documented South Asian pedagogical expertise, while the user

expects systematic divergence on the most-critical 'providing guidance' dimension between this pool and Indian professional teachers. A MODERATE-priority concern attaches to Output Ontology: equal-weight DAMR does not reflect the user's prioritization of 'providing guidance,' and that dimension's rubric implicitly favors Socratic scaffolding over the direct-correction norms valued by Indian teachers. Field-level evidence (MathTutorBench, Shiksha Copilot study, 2025 ITS survey) confirms these are not local artifacts but field-wide gaps in pedagogically- and cross-culturally-grounded benchmarking.

MATHDIAL (risk: HIGH)

MathDial presents a structurally sound but culturally narrow evaluation framework for AI tutoring quality. For the Indian Metropolitan Professional Teachers (Grades 1–8) deployment, three dimensions are well-aligned (Input Form, Output Form, and — given user confirmation that the eight categories suffice — Input Ontology), while three dimensions raise serious concerns. Input Content (HIGH user priority) shows fundamental misalignment: the LLM-simulated student exhibits verbose, peer-register, reasoning-out-loud behavior diametrically opposed to the deferential brief low-verbalization style the user identifies as characteristic of Indian metro classrooms, math word problems are entirely US-origin, and LLM looping artifacts produce unnatural student behavior. Output Content (HIGH user priority) is the deepest concern: ground-truth labels were produced by predominantly UK/US annotators explicitly instructed toward Socratic non-directive scaffolding, with $\kappa=0.34$ inter-annotator agreement on Equitable Tutoring even within the Western pool — and the user explicitly expects Indian professional teachers' judgments to systematically diverge on direct-correction vs. probing and exam-readiness weighting. Output Ontology (MODERATE) embeds a constructivist anti-telling structural orientation (Telling@k as negative metric, Telling as last-resort) that conflicts with Indian professional teachers' view of direct correction as a primary pedagogical tool, and the user's prioritization of 'providing guidance' is not commensurately weighted in the three co-equal evaluation dimensions. The English text-only modality match is the benchmark's strongest validity asset for this deployment.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 4/5 (Minor gaps) [high confidence]
The user explicitly accepted curriculum-agnostic math errors as sufficient and confirmed the eight pedagogical dimensions are necessary and sufficient (elicitation [Q1, Q3], reducing concern at the taxonomy level. The benchmark covers two difficulty levels (elementary and middle-school) [Q143, Q144], aligning with Grade 1–8 scope. The taxonomy is curriculum-agnostic by construction [Q117, Q118] and was validated against learning-science principles with a pilot confirming necessity and sufficiency [Q129, Q131, Q132]. Minor concerns: the taxonomy reflects a Socratic/answer-withholding pedagogical model [Q17, Q23] that the deployment context flags as partially conflicting with Indian direct-correction norms, but this is more an output-rubric concern (handled in OO/OC) than an input-category gap. No Indian curriculum-specific subcategories (e.g., NCERT topics, lakh/crore conventions) are present, but the user explicitly stated these are not required.	The user explicitly confirmed that curriculum-agnostic math errors and the benchmark's eight pedagogical categories are necessary and sufficient for the deployment task [A1, A3], substantially reducing ontological misalignment risk. The taxonomy is theoretically grounded in Reiser's (2004) scaffolding framework distinguishing structure (Focus) from problematize (Probing) moves [Q36, Q37], with Telling and Generic as additional categories [Q40, Q41]. The dataset confirms transparent application of the four-move taxonomy across all sampled dialogues, providing structured pedagogical vocabulary [DATASET-D2, DATASET-D3]. Confusion type coverage spans six algebra-misconception categories [DATASET-D7, DATASET-D8, DATASET-D10]. Minor concerns remain: the taxonomy's structural privileging of scaffolding over directive correction [Q154, Q155] embeds a constructivist orientation that, while not omitting categories the user requires, frames the relationships among them in a way that may not match Indian metro teaching priorities. The benchmark also explicitly omits meta-cognitive support and rapport-building [Q112], though the user did not flag these as required.
MRBENCH evidence	MATHDIAL evidence

[Q117] 'The proposed taxonomy primarily focuses on the task of the student mistake remediation in the domain of mathematics.' (p.9)	[Q36] 'We base the first two on the work of Reiser (2004), who suggest that scaffolding strategies can be split into two main categories: structure and problematize.' (p.5)
[Q129] 'Through an iterative analysis of the taxonomy, we identify eight dimensions that comprehensively assess tutor response quality in the context of mistake remediation.' (p.12)	[Q112] 'Inspired by previous work in scaffolding, we acknowledge our focus is on a subset of common teaching moves. However, this does not cover all the goals of human tutors, such as meta-cognitive support or building rapport with a student.' (p.10)
[Q132] 'our validation pilot study (see Section 4.2) confirmed that the selected eight dimensions are both necessary and sufficient for evaluating tutor response quality in dialogues aimed at mistake remediation.' (p.12)	[Q154] 'Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing.' (p.17)
[Q143] 'including both datasets in MRBench ensures diversity and provides for a good mix of easy and difficult mathematical problems' (p.14)	[Q155] 'On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck.' (p.17)
[Q144] 'The problems covered in the Bridge dataset are at the elementary school level, whereas those in MathDial are at the middle school level.' (p.14)	DATASET-D2: Six consecutive (probing) labels in one dialogue — taxonomy systematically applied at utterance level
	DATASET-D3: Telling moves explicitly labeled and visible in transcript
	DATASET-D7: Procedural confusion student profile category represented
	DATASET-D8: Underlying-principle confusion category represented
	DATASET-D10: Question-misunderstanding confusion category represented

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

MRBENCH — 2/5 (Significant gaps) [high confidence]	MATHDIAL — 1/5 (Serious concern) [high confidence]
This is a HIGH-priority dimension per elicitation. The benchmark content is drawn from Bridge [Q44–Q47] and MathDial [Q48, Q49], both Western-academic English-language datasets. The user explicitly confirmed (elicitation Q2) meaningful mismatches in student communication style: Indian Grade 1–8 students are briefer, more deferential, and procedure-focused, while MathDial features extended multi-turn structured dialogues [Q95]. The Bridge selection criterion centers on student last-utterance error [Q47] but does not document misconception types relevant to Indian rote-procedural traditions, L1-interference errors, or NCERT-specific algorithmic conventions. No documentation acknowledges cultural provenance as a limitation. MathDial's student is an LLM, not a real student [Q48], introducing an additional layer of cultural distance from authentic Indian student behavior. Field-level evidence confirms this is not a local artifact: no Indian-classroom-grounded math tutoring benchmark exists [WEB-16, WEB-19].	Input content is the highest-priority concern (user-flagged HIGH) and the dataset analysis confirms severe construct-irrelevant variance. The student simulation is calibrated to Western tutoring conventions [Q32, Q34] with documented limitations in representing human learning [Q106]. Empirically, the LLM-simulated students invariably produce verbose, multi-sentence, reasoning-out-loud responses [DATASET-D6, DATASET-D7, DATASET-D9] — the polar opposite of the brief, deferential, low-verbalization style the user identifies as characteristic of Indian metro classrooms [A2]. Students even use casual peer-register openers ('Hi, I'm doing well. How can I help you?') [DATASET-D24] inconsistent with deferential Indian student-teacher interactions. Math word problems are entirely US-origin from GSM8k [Q103, Q118] with US currency and scenarios (recycling deposits, temp agencies) [DATASET-D15, DATASET-D16, DATASET-D17] and no Indian cultural grounding. Nominal name diversity [Q123, DATASET-D11] does not extend to behavioral diversity. Additional concern: simulated student loops on identical wrong answers across multiple teacher corrections [DATASET-D13, DATASET-D14] — an LLM artifact not representative of any real student population. Authors explicitly acknowledge teachers may have adapted to model patterns rather than exhibiting naturalistic behavior [Q108]. The MathDial dialogues thus encode interaction dynamics fundamentally misaligned with Indian metro classrooms at the level of every individual datapoint.
MRBENCH evidence	MATHDIAL evidence
[Q44] 'The Bridge dataset (Wang et al., 2024a) comprises partial dialogue interactions between real human tutors and students at the elementary level' (p.5)	[Q32] 'We use InstructGPT (text-davinci-003) to generate student turns.' (p.5)
[Q48] 'The dialogues in the MathDial dataset (Macina et al., 2023) consist of complete multi-turn conversations between a real human tutor and an LLM acting as a student' (p.5)	[Q103] 'Our dataset consists of ca. 3k tutoring conversations grounded in math word problems from GSM8k.' (p.9)
[Q95] 'In contrast, for MathDial, the contexts are longer, the mistakes are grounded in reasoning, and the responses are more structured.' (p.8)	[Q106] 'In this work, we used an LLM to simulate student confusion. However, we acknowledge that these models have a limited understanding of human learning and this is a key limitation in our dataset – certain kinds of student confusions may be under- or over-represented in our dataset.' (p.9)
[Q47] 'the key one was that the student's last utterance (or last few utterances) should exhibit an error or confusion.' (p.5)	[Q108] 'Furthermore, in our setup, teachers were interacting with an LLM role-playing as a student. However, it is possible that some teachers might have learned to interact with the student model in a different way than they would do in the classroom.' (p.10)
[WEB-16] MathTutorBench (EMNLP 2025) is also Western-sourced and operationalizes good tutoring as Socratic withholding, confirming field-wide cultural provenance gap	[Q118] 'While the problems in GSM8k are simple enough to be understood quickly by teachers, they remain challenging for students, who among others have to deal with equations or percentages.' (p.14)

[WEB-19] 2025 ITS evaluation survey identifies absence of pedagogy-driven, cross-cultural evaluation frameworks; no India-specific ITS benchmark cited	[Q123] 'To build a dataset that would reflect students of various backgrounds, we use numerous student names associated with their given pronouns.' (p.15)
	DATASET-D6 : Simulated student gives unprompted multi-sentence reasoning walkthrough — opposite of deferential brief Indian student style
	DATASET-D7 : Student volunteers full reasoning rationale unsolicited
	DATASET-D9 : Three-step unprompted reasoning walkthrough by simulated student
	DATASET-D13 : Student verbally agrees with correction while repeating wrong calculation — unnatural LLM artifact
	DATASET-D14 : Identical wrong statement repeated three times across teacher moves — LLM looping unrepresentative of real students
	DATASET-D15 : US dollar amounts and US-style names (Cecil, Catherine, Carmela)
	DATASET-D16 : US bottle-deposit recycling system with no Indian analogue
	DATASET-D17 : US temp-agency labor market context
	DATASET-D24 : Casual peer-register opener 'Hi, I'm doing well. How can I help you?' — not deferential Indian student register
	DATASET-D11 : South Asian student name (Rishi) but Western-expressive behavior pattern unchanged

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 5/5 (Strong alignment) [high confidence]
The benchmark is text-only English [Q133, Q134, Q147], matching the deployment's text-based English mobile/enterprise application interface. The target population is fluent in English professionally. No script mismatch, no multimodal mismatch, and adequate infrastructure (urban Tier-1 internet penetration ~80–88%, smartphone access near-universal for professional teachers) [WEB-5, WEB-6, WEB-7] support text-based English signal delivery. Minor concern: no multimodal capture of handwritten student work, which is common in actual Indian classrooms, but the evaluator-role deployment (teachers judging text dialogues) does not require this. The user weighted IF as LOWER priority.	MathDial is English text-only [Q1, Q32], and the deployment context is confirmed English text-only on mobile/enterprise platforms with no code-switching, script variation, or vernacular content. There is no modality, script, or language-encoding mismatch. Dataset analysis confirms all 18 sampled examples are pure English prose with no Hindi, Hinglish, or transliteration [DATASET-D6, DATASET-D15]. A safety filter using Perspective API removes toxic content [Q157, Q158]. Minor surface-level imperfections in teacher utterances (spelling/typing errors) [DATASET-D19] are noted but do not affect form validity. The deployment is Indian metro with English-medium instruction, and metro infrastructure supports the input modality [WEB-3, WEB-4, WEB-8].
MRBENCH evidence	MATHDIAL evidence
[Q133] 'The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2.' (p.12)	[Q1] 'We describe how we use this framework to collect MATHDIAL, a dataset of 3k one-to-one teacher-student tutoring dialogues grounded in multi-step math reasoning problems.' (p.1)

[Q147] 'In all cases, length is estimated using the number of characters.' (p.14)	[Q32] 'We use InstructGPT (text-davinci-003) to generate student turns.' (p.5)
[WEB-5] Urban Tier-1 internet penetration ~88%, broadband >50% in urban households	[Q157] 'As we are interested in real educational use cases for our tutoring system, we apply a safety filter to filter out conversations with any sensitive content.' (p.18)
[WEB-6] India had 660 million smartphone users as of 2024–2025	[Q158] 'In particular, we use the Perspective API to filter out conversations containing toxic content (<1%).' (p.18)
[WEB-7] National mobile data median speed 94.62 Mbps in early 2024	[WEB-3] Delhi has 3,000+ English-medium schools
	[WEB-4] Delhi government Schools of Excellence operate in English medium
	[WEB-8] National school internet connectivity 63.5% per UDISE+ 2024-25
	DATASET-D6: All turns in standard English prose; no Hindi/Hinglish/transliteration
	DATASET-D15: Question, ground truth, student solution, conversation all in English
	DATASET-D19: Minor spelling errors in teacher utterances do not affect form validity

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

MRBENCH — 3/5 (Moderate gaps) [high confidence]	MATHDIAL — 2/5 (Significant gaps) [high confidence]
MODERATE priority per elicitation. The eight dimensions and three-tier labeling [Q58, Q59] are accepted by the user as necessary and sufficient (elicitation Q3). However, two structural concerns affect deployment validity. First, DAMR weights all eight dimensions equally [Q64], whereas the user identified 'providing guidance' as the single most critical dimension; using DAMR directly as an acceptability proxy would dilute the signal Indian teachers care most about. Second, the 'providing guidance' rubric operationalizes guidance via Socratic scaffolding (hints, supporting questions, not revealing answers) [Q23, Q24], which the user (elicitation Q4) and field evidence [WEB-15, WEB-16] indicate may not match Indian direct-correction expectations — a structural validity concern about how the construct itself is operationalized. The decision rule for the most-critical dimension may encode a non-regional pedagogical value. Treating dimensions as orthogonal [Q30] is acknowledged as a simplification [Q116, Q120, Q121]. Tone label collapse to 'Non-offensive' [Q90] reaches 100% compliance, providing little discriminative signal.	User-flagged MODERATE risk. The user accepted the eight categories as necessary and sufficient [A3], reducing one source of ontological gap. However, the benchmark's output taxonomy operationalizes Equitable Tutoring [Q97, Q98, Q166] with an explicitly anti-telling, constructivist orientation, treating Telling as a last resort [Q155] and using Telling@k as a negative metric [Q96] — directly conflicting with the Indian metro pedagogical norm of direct correction as a primary, not last-resort, strategy. The user identified 'providing guidance' as the single most critical dimension [A3], but the benchmark distributes evaluation across three co-equal dimensions (Coherence, Correctness, Equitable Tutoring) [Q97, Q166], not weighting guidance commensurately. Dataset analysis confirms structural embedding of these values: Telling is positioned as late-stage fallback after multiple Focus/Probing failures [DATASET-D3, DATASET-D5, DATASET-D27], and self-correctness annotations frame Telling as 'had to reveal' [DATASET-D5, DATASET-D23] — a quality-deficit framing rather than legitimate pedagogical strategy. Additionally, the benchmark's classification of teacher openings that fully restate correct setups as 'Generic' rather than 'Telling' [DATASET-D12, DATASET-D28] reveals taxonomy ambiguity Indian annotators might resolve differently. This is structural validity violation: the construct's structure (what counts as good guidance) is misrepresented for the target context.
MRBENCH evidence	MATHDIAL evidence

[Q58] 'Each dimension was annotated using a three-tier labeling system' (p.6)	[Q96] 'Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns.' (p.7)
[Q64] 'Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels.' (p.6)	[Q97] 'Finally, we conduct a human evaluation according to three criteria: 1) Coherence... 2) Correctness... and 3) Equitable tutoring.' (p.8)
[Q23] 'a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance.' (p.4)	[Q98] 'Equitable tutoring describes how well the model provides the student with room for exploring the problem and solution space.' (p.8)
[Q24] 'a good tutor response should provide the student with relevant and helpful guidance, such as a hint, an explanation, or a supporting question.' (p.4)	[Q155] 'On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck.' (p.17)
[Q30] 'we explicitly instructed all annotators to treat each dimension as independent and orthogonal to minimize confounding factors' (p.4)	[Q166] 'Equitable tutoring - The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways.' (p.18)
[Q90] 'When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans.' (p.8)	DATASET-D3: Focus moves deployed multiple times before Telling — annotator instruction to delay Telling structurally embedded
[Q120] 'one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning.' (p.9)	DATASET-D5: Self-correctness 'Yes, but I had to reveal the answer' — Telling framed as deficit even when successful
[WEB-15] Indian teachers value direct, curriculum-aligned outputs over Socratic scaffolding; misalignment with Western-default AI outputs documented	DATASET-D12: Teacher recap of correct setup labeled 'generic' not 'telling' — taxonomy ambiguity
[WEB-16] MathTutorBench reward model also operationalizes good tutoring as Socratic withholding, confirming field-wide rubric default	DATASET-D23: Three consecutive Telling moves labeled 'had to reveal' — deficit framing of successful direct instruction
	DATASET-D27: Single Telling resolves stuck conversation, framed as fallback
	DATASET-D28: Opening with full correct setup labeled 'generic' — classification Indian annotators may resolve differently

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

MRBENCH — 2/5 (Significant gaps) [high confidence]	MATHDIAL — 1/5 (Serious concern) [high confidence]
HIGH priority per elicitation. The annotation pool is four MBZUAI CS post-graduates [Q31] explicitly without required teaching experience [Q32], with no documented South Asian educational expertise. The user expects systematic divergence (elicitation Q4) between Indian teachers' guidance-quality judgments and this annotator pool's, particularly on direct correction vs. Socratic probing, exam-readiness framing, and teacher-directed pedagogy. Field evidence corroborates: Indian teachers structurally value direct, curriculum-aligned outputs over Western-default Socratic scaffolding [WEB-15]. Inter-annotator agreement (Cohen's $\kappa=0.71$, Fleiss' $\kappa=0.65$) [Q39, Q61] demonstrates internal consistency within the MBZUAI pool but provides no evidence of convergent validity with Indian teachers — the gold standard [Q68] reflects MBZUAI annotators' implicit pedagogical model. LLM-critic correlations were predominantly negative [Q107], confirming pedagogical judgments are not easily automatable. The most-critical dimension's labels were calibrated by annotators whose pedagogical priors likely diverge from Indian professional teachers'. No India-specific annotation study exists [WEB-19].	User-flagged HIGH risk and the highest-priority validity concern. Ground-truth quality judgments were produced by 91 professional teachers recruited via Prolific [Q23], predominantly UK nationals, followed by USA, Canada, Australia, India, Germany [Q27] — India is the fifth of six named countries with no disaggregated count reported. Annotators were explicitly instructed not to correct student solutions by telling but to give students opportunities to explore [Q148], systematically encoding Western constructivist norms regardless of annotator origin. Inter-annotator agreement is concerning: $\kappa=0.60$ between independent annotators but only $\kappa=0.49$ and $\kappa=0.34$ between annotators and original teacher for move labeling [Q69], and critically $\kappa=0.34$ for Equitable Tutoring [Q100] — the dimension most sensitive to cross-cultural variation. The user explicitly expects systematic divergence between Indian professional teachers' judgments and the benchmark's annotator-derived ground truth, particularly on direct-correction vs. Socratic probing and exam-readiness weighting [A4]. Empirically, the dataset shows pervasive Socratic-mode dialogues with multiple Focus/Probing moves before any Telling [DATASET-D2, DATASET-D3], gentle non-directive corrections [DATASET-D20], and 'had to reveal' framing on successful Telling resolutions [DATASET-D5, DATASET-D23] — patterns the user identifies as misaligned with Indian professional norms. Web research corroborates: PARIKSHA found LLM evaluators align less with human evaluations on culturally nuanced responses [WEB-18], and systematic review confirms majority Indic datasets are translations with poor socio-cultural alignment [WEB-15]. No India-specific math tutoring dialogue benchmark exists [WEB-15, WEB-16, WEB-17, WEB-18]. This violates both convergent validity (labels do not correlate with regional perspectives) and external validity (judgments do not generalize).
MRBENCH evidence	MATHDIAL evidence
[Q31] 'The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.' (p.5)	[Q23] 'We recruit professionals with teaching experience through Prolific. We only select teachers who have completed at least 500 submissions and achieved a 100% completion rate.' (p.4)
[Q32] 'we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.' (p.5)	[Q27] 'The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years.' (p.4)
[Q61] 'The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement' (p.6)	[Q69] 'We obtain an agreement of $\kappa = 0.60$ between the two annotators and $\kappa = 0.49$ and $\kappa = 0.34$, respectively, between either of the annotators and the teacher.' (p.6)

[Q68] 'We consider human-based evaluations as gold standard.' (p.7)	[Q100] 'We obtain agreements of $\kappa = 0.29$, $\kappa = 0.69$, and $\kappa = 0.34$ for the three categories.' (p.8)
[Q107] 'Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions.' (p.8)	[Q148] 'Specifically, they were instructed to not just correct student solutions by telling what's correct/incorrect, but to give students the opportunity to explore the problem with a focus on core aspects, such as their chosen strategy.' (p.17)
[WEB-15] Indian teachers face structural misalignment with Western-default AI outputs; value curriculum-aligned, direct guidance	[WEB-15] Systematic review: majority Indic LLM datasets are translations and models fare poorly on socio-cultural tasks
[WEB-19] No India-specific or cross-cultural ITS evaluation framework cited in 2025 survey	[WEB-18] PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses
	[WEB-16] IndicEval covers Indian high-stakes exams but is not a pedagogical dialogue benchmark
	[WEB-17] Hindi LLM benchmarks suite focuses on math reasoning not tutoring quality
	DATASET-D2: Six consecutive probing questions before student arrives at answer — high Equitable Tutoring score by MATHDIAL standards, likely under-directive by Indian standards
	DATASET-D3: Focus moves deployed four times before Telling — annotator instruction visibly embedded
	DATASET-D5: 'Yes, but I had to reveal the answer' framing of successful Telling resolution
	DATASET-D20: Vague gentle correction without identifying specific error
	DATASET-D23: Three Telling moves successfully resolve confusion but framed as 'had to reveal'

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 5/5 (Strong alignment) [high confidence]
LOWER priority per elicitation. Output form (categorical labels [Q58, Q59], DAMR percentages, and Pearson correlations [Q64, Q66]) is appropriate for the deployment scenario where teachers render acceptability judgments on AI+human-augmented responses. The benchmark explicitly avoids unreliable NLG metrics like BLEU/BERTScore [Q4, Q7, Q8] in favor of pedagogically meaningful categorical scores. No mismatch with deployment output needs is identified. Minor concern: the benchmark reports per-dimension DAMR but no aggregate acceptability score weighted by deployment priorities; teams would need to compute their own weighted aggregate to map to a binary acceptable/unacceptable judgment, but this is straightforward.	Output form is well-aligned. Both benchmark and deployment evaluate text-based open-ended tutoring dialogue outputs. Automatic metrics (sBLEU, BERTScore, KF1, Uptake) [Q89, Q90, Q93] and interactive metrics (Success@k, Telling@k) [Q95, Q96] operate on text. Human evaluation uses 3-point Likert and binary scales [Q167] in standard formats. Authors transparently acknowledge automatic faithfulness metrics are insufficient [Q91, Q92] and that immediate solving success does not capture long-term learning [Q114]. The deployment context (English text-only on mobile/enterprise platforms) matches output modality completely. No literacy or accessibility issues for the target population (English-fluent professional teachers per population characteristics).
MRBENCH evidence	MATHDIAL evidence

[Q58] 'Each dimension was annotated using a three-tier labeling system' (p.6)	[Q89] 'We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ($uT + 1$) and annotated response ($u^*T + 1$) for each teacher response in the conversation.' (p.7)
[Q64] 'Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels.' (p.6)	[Q91] 'However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable.' (p.7)
[Q66] 'Annotation Correlation (AC): This metric is based on Pearson's correlation' (p.6)	[Q114] 'Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired.' (p.10)
[Q4] 'General domain-agnostic natural language generation (NLG) metrics ... are not well-suited for this context, as most of them fail to account for pedagogical values' (p.1)	[Q167] 'We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness.' (p.18)